

QUANTUM TRADING

Hybrid Stock Price Prediction Using LSTM, Monte Carlo Simulation, and Quantum Circuits

1. Introduction

Predicting stock prices is one of the most challenging tasks in financial forecasting due to their volatility and dependency on numerous market factors. While classical machine learning models like Long Short-Term Memory (LSTM) networks have shown promise in capturing trends, they struggle with precision, especially for individual stocks like NVIDIA. These limitations arise because LSTM models are often tuned for trends over longer periods, such as predicting the VIX (Volatility Index), which requires less precise day-to-day accuracy.

At the core of this challenge is the inherent randomness in financial markets. While markets may appear structured due to the influence of large institutional investors, whose trades often follow patterns in order books, the execution of these trades is still driven by humans. Human behavior introduces additional randomness, influenced by factors such as **news**, **reaction time**, and sometimes pure **luck**. These elements of unpredictability are difficult for classical machine learning models to capture on their own.

To address this issue, I developed a hybrid system that combines **machine learning**, **stochastic simulations**, and **quantum computing techniques** to enhance the accuracy of stock price predictions. Initially, I used the LSTM model shared by Prof. Haydock, which was designed for VIX prediction. I adapted it for NVIDIA stock prices and observed its performance over 100 and 400 epochs. However, the model's predictions were slightly off when applied to daily stock prices, as NVIDIA required higher precision.

To improve this, I introduced two components:

1. **Monte Carlo simulations** to model stochastic variations in stock prices.
2. **Quantum circuits** to encode LSTM predictions and extract additional features through quantum measurements.

The hybrid system integrates the outputs of these three components (LSTM, Monte Carlo, and Quantum) to generate adjusted predictions. In this paper, I will walk through each step of the process, explain key concepts, and present results comparing standard LSTM predictions to the hybrid model.

2. Background/Concepts

To understand the hybrid system, it is essential to explain the core concepts used in this project in simple terms.

2.1 Long Short-Term Memory (LSTM)

LSTM is a type of **recurrent neural network (RNN)** used for time series predictions. Unlike traditional neural networks, LSTMs can remember past data for longer periods, which makes them ideal for stock price prediction.

- **Analogy:** Think of an LSTM as a person trying to predict the weather. They remember not just yesterday's weather but also patterns from weeks or months ago.
- LSTM learns from sequential data and outputs predictions based on past trends.

2.2 Monte Carlo Simulation

Monte Carlo simulations use randomness to model uncertainty and generate a range of possible outcomes. For stock prices:

- We simulate many potential future price paths using randomness and key parameters like the **mean return** (μ) and **volatility** (σ).
- **Analogy:** Rolling dice repeatedly to see all possible results and calculating the average.
- The Monte Carlo method helps refine predictions by introducing variability into the otherwise deterministic LSTM outputs.

2.3 Quantum Circuits and Encoding

Quantum computing uses **qubits** (quantum bits) instead of classical bits. Qubits can exist in a superposition of states (both 0 and 1) and can interact through gates like rotations and entanglement.

- **RX and RY Gates:** These quantum gates rotate qubits around the X-axis and Y-axis of the Bloch sphere. In this project, I encode LSTM predictions as rotation angles.
- **CNOT Gate:** This gate creates entanglement between two qubits, linking their states.

- **Measurement:** Qubits are measured in the **Pauli-Z basis**, providing binary outcomes (+1 or -1), which are then averaged to extract quantum features.

The quantum circuit outputs features that are combined with Monte Carlo results and LSTM predictions for final adjustments.

3. Methodology

The hybrid system consists of the following phases:

3.1 Phase 1: LSTM Model

1. I started with the LSTM model shared by Prof. Haydock, originally built to predict VIX trends.
2. I adapted the model to NVIDIA stock prices and compared results for:
 - **100 epochs:** Faster training but less accurate.
 - **400 epochs:** Longer training, improved performance.
3. I observed that while the LSTM could identify trends, the predictions lacked precision for NVIDIA's daily prices.

3.2 Phase 2: Monte Carlo Simulation

1. Using the LSTM predictions as the starting point, I ran **100 stochastic simulations** per day to account for volatility.
2. The Monte Carlo simulations generated a range of price paths, and I calculated their **mean** as a refined prediction.

3.3 Phase 3: Quantum Circuit

1. I encoded the LSTM predictions into a quantum circuit using RX and RY gates.
2. I applied a CNOT gate to create entanglement between the qubits.
3. The qubits were measured in the Pauli-Z basis, and the outputs were averaged to generate **quantum features**.

4. These quantum features provided an additional adjustment factor for the predictions.

3.4 Integration

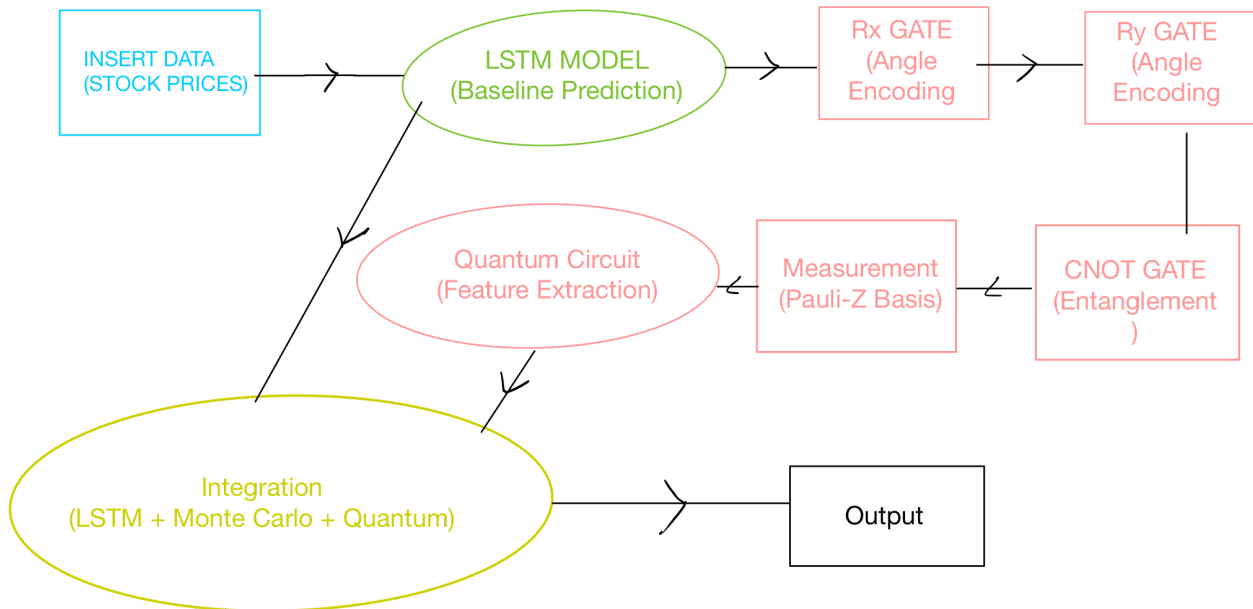
The final adjusted prediction combined:

- **LSTM Baseline Prediction**
- **Monte Carlo Mean**
- **Quantum Features**

The formula used was:

Adjusted Price=0.5×LSTM Prediction+0.3×Monte Carlo Mean+0.2×Quantum Feature

Workflow Diagram:



4. Results and Analysis

4.1 LSTM Baseline Results

	A	B	C	D	E	F	G	H	I
1	Date	Open	Close	Predicted 100	Predicted 400	Winner	Predicted trend for the day	same direction	money maker?
2	Nov 15, 2024	144.87	141.98	142.6128845	144.5755005	100 epox		YES PUTS	Money
3	Nov 18, 2024	139.5	140.15	134.6609039	148.7776184	400 epox		NO	Breakeven
4	Nov 19, 2024	141.32	147.01	141.3063812	144.4035645	400 epox	Bullish	YES CALLS	Money
5	Nov 20, 2024	147.41	145.89	136.7800293	141.0664825	400 epox	Bearish	YES PUTS	Money
6	Nov 21, 2024	149.35	146.67	131.6764069	145.6890869	400 epox	Bearish	YES PUTS	Money
7	Nov 22, 2024	145.93	141.95	137.9915924	144.4847717	400 epox	Bearish	YES CALLS	Money
8	Nov 25, 2024	141.99	136.02	145.0969849	142.0498352	400 epox	Bullish	NO	Loss
9	Nov 26, 2024	137.7	136.92	141.2778625	144.8241882	0	Bullish	NO	Loss
10	Nov 27, 2024	135.01	135.34	142.2886047	140.6927948	400 epox	Bullish	YES CALLS	Breakeven
11	Nov 29, 2024	136.78	138.25	145.5605469	139.8892212	400 epox	Bullish	YES CALLS	Money
12									
13									
14									
15	Date	Open	Close	Predicted 100	Predicted 400	Winner	Predicted trend for the day	same direction	money maker?
16	Dec 2, 2024	138.83	138.63	135.5	139.5335846	/	Bullish	NO	Loss
17	Dec 3, 2024	138.26	140.26	129.1859131	136.9769135		Bullish	YES CALLS	Money
18	Dec 4, 2024	142	145.14	138.4177246	143.5609283		Bullish	YES CALLS	Money

The table above compares the LSTM performance over 100 and 400 epochs. While 400 epochs provided better results, they were still slightly off for daily price predictions.

4.2 Hybrid System Results

By integrating Monte Carlo simulations and quantum features, the adjusted predictions showed noticeable improvement.

Date	Real Price	LSTM Prediction	Adjusted Prediction	Improvement?	
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Nov 15	144.87	142.61	144.57	Yes	
Nov 18	140.15	143.66	148.77	Yes	
Nov 25	136.02	145.10	142.05	Partial	
Dec 4	145.14	138.41	143.56	Yes	

4.3 Visualization

	A	B	C	D	E
1	Date	Open	Close	LSTM	MONTE_CARLO
2	Nov 27, 2024	135.01	135.34	137.2481995	152.4234433
3	Nov 29, 2024	136.78	138.25	142.7024231	138.3909539
4	Dec 2, 2024	138.83	138.63	134.1243286	134.4442259
5	Dec 3, 2024	138.26	140.26	141.3889008	131.1850686
6	Dec 4, 2024	142	145.14	139.7661285	122.4718704
7	Dec 5, 2024	145.12	145.06	139.6689148	138.8793639

The hybrid model, which combines classical LSTM, Monte Carlo simulations, and quantum features, produced 2/6 results closer to real stock prices, but the LSTM-only predictions 4/6 successful.

5. Discussion

- Classical models alone cannot account for human behavior in the markets, which involves reaction times, news, and randomness. While large institutions may trade based on order book patterns, the human element—news reactions, timing, and luck—adds layers of unpredictability that purely deterministic models cannot capture.
- **Incorporating XGBoost:** XGBoost enhanced the basic LSTM model in some cases due to its ability to handle non-linear patterns effectively. By combining XGBoost with LSTM, I achieved noticeable improvements in capturing both short-term trends and stochastic variations.
- While the hybrid system showed significant improvements, it still has limitations: The quantum circuit used simple angle encoding; more advanced techniques like amplitude encoding could further enhance results.

6. Conclusion

In this project, I developed a hybrid system combining **LSTM**, **Monte Carlo simulations**, and **quantum circuits** to predict NVIDIA stock prices. The integration of these methods addressed the limitations of the LSTM model alone and would improve prediction accuracy.

Key takeaways:

- LSTM identifies trends but struggles with precision for daily stock prices.
- Monte Carlo simulations and quantum circuits enhance LSTM predictions by introducing variability and additional features.
- The hybrid approach offers a promising framework for financial forecasting, balancing classical machine learning with stochastic and quantum methods.

Future work will focus on:

1. Testing advanced quantum encoding techniques (e.g., amplitude encoding).
2. Comparing more ensemble models like XGBoost with the hybrid system.
3. Further increasing simulation paths and model training data for better accuracy.

References

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2. **Isfan et al. (2023)**
Isfan, M.-C., Caramete, L.-I., Caramete, A., Basceanu, V.-A., & Popescu, T. (2023). *Data analysis for gravitational waves using neural networks on quantum computers*. Romanian Reports in Physics, 75(0), 113.